PySpark Capstone Project 2

Smart Retail Inventory and Sales Analytics

Business Scenario

A retail chain wants to analyze:

```
Stores
Products
Inventory
Sales
Suppliers
Revenue
Stock Availability
Reorder Risk
Store Performance
Category Performance
```

The goal is to build a complete PySpark ETL pipeline using CSV and JSON files.

Dataset 1: stores.csv

```
%%writefile stores.csv
store_id,store_name,city,state,store_type,manager_name
S101,Metro Mart Hyderabad,Hyderabad,Telangana,Supermarket,Rahul Sharma
S102,Metro Mart Bangalore,Bangalore,Karnataka,Supermarket,Priya Reddy
S103,Metro Mart Mumbai,Mumbai,Maharashtra,Hypermarket,Amit Kumar
S104,Metro Mart Chennai,Chennai,Tamil Nadu,Supermarket,Sneha Patel
S105,Metro Mart Delhi,Delhi,Delhi,Hypermarket,Farhan Ali
S106,Metro Mart Pune,Pune,Maharashtra,Mini Store,Neha Singh
S107,Metro Mart Kochi,Kochi,Kerala,Mini Store,Arjun Verma
S108,Metro Mart Jaipur,Jaipur,Rajasthan,Supermarket,Meera Nair
```

Dataset 2: products.csv

```
%%writefile products.csv
product_id,product_name,category,brand,supplier_id,unit_price
P101,Laptop,Electronics,Lenovo,S201,65000
P102,Mobile,Electronics,Samsung,S202,25000
P103,Television,Electronics,LG,S203,45000
P104,Office Chair,Furniture,Featherlite,S204,7000
P105,Study Table,Furniture,Urban Ladder,S204,12000
P106,Shoes,Fashion,Nike,S205,4500
P107,Watch,Fashion,Fastrack,S206,8000
P108,Backpack,Fashion,Wildcraft,S206,2500
P109,Refrigerator,Electronics,Whirlpool,S203,38000
P110,Sofa,Furniture,Godrej,S204,32000
P111,Headphones,Electronics,Sony,S999,3000
P112,T-Shirt,Fashion,Puma,,1500
```

Dataset 3: inventory.csv

```
%%writefile inventory.csv
inventory_id,store_id,product_id,stock_quantity,reorder_level,last_update
I1001,S101,P101,10,5,2026-01-10
I1002,S101,P102,25,10,2026-01-10
I1003,S101,P104,3,5,2026-01-11
I1004,S102,P101,8,5,2026-01-12
I1005,S102,P103,5,4,2026-01-12
I1006,S103,P105,2,5,2026-01-13
I1007,S103,P106,30,10,2026-01-14
I1008,S104,P107,4,5,2026-01-15
I1009,S105,P108,50,20,2026-01-15
I1010,S106,P109,,6,2026-01-16
I1011,S107,P110,1,3,2026-01-17
I1012,S108,P120,12,5,2026-01-18
```

Dataset 4: sales.csv

```
%%writefile sales.csv
sale_id,store_id,product_id,sale_date,quantity_sold,sale_amount,payment_m
SA1001,S101,P101,2026-01-10,1,65000,UPI
SA1002,S101,P102,2026-01-10,2,50000,Card
SA1003,S102,P101,2026-01-11,1,65000,UPI
SA1004,S103,P106,2026-01-12,4,18000,Cash
SA1005,S104,P107,2026-01-12,1,8000,Card
SA1006,S105,P108,2026-01-13,5,12500,UPI
SA1007,S106,P109,2026-01-14,1,38000,Card
SA1008,S107,P110,2026-01-15,1,32000,UPI
SA1009,S108,P120,2026-01-15,2,10000,Cash
SA1010,S101,P104,2026-01-16,2,14000,
SA1011,S102,P103,2026-01-17,1,,UPI
SA1012,S103,P105,2026-01-18,1,12000,Card
SA1013,S104,P107,2026-02-01,2,16000,UPI
SA1014,S105,P108,2026-02-02,3,7500,Cash
SA1015,S101,P102,2026-02-03,1,25000,Card
```

Dataset 5: suppliers.json

```
%%writefile suppliers.json
[
  {
    "supplier_id": "S201",
    "supplier_name": "TechSource India",
    "city": "Hyderabad",
    "rating": 4.5,
    "contact": {
      "phone": "9876500011",
      "email": "techsource@mail.com"
    }
  },
  {
    "supplier_id": "S202",
    "supplier_name": "MobileWorld Distributors",
    "city": "Bangalore",
    "rating": 4.2,
    "contact": {
      "phone": null,
      "email": "mobileworld@mail.com"
```

```
    }
  },
  {
    "supplier_id": "S203",
    "supplier_name": "HomeTech Supply",
    "city": "Mumbai",
    "rating": 4.4,
    "contact": {
      "phone": "9876500013",
      "email": null
    }
  },
  {
    "supplier_id": "S204",
    "supplier_name": "Urban Furniture Co",
    "city": "Delhi",
    "rating": 4.0,
    "contact": {
      "phone": "9876500014",
      "email": "urban@mail.com"
    }
  },
  {
    "supplier_id": "S205",
    "supplier_name": "Fashion Direct",
    "city": "Pune",
    "rating": 3.8,
    "contact": {
      "phone": null,
      "email": null
    }
  }
]
```

Capstone Tasks

Part 1: Ingestion

```
1. Read stores.csv.
2. Read products.csv.
```

3. Read inventory.csv.
4. Read sales.csv.
5. Read suppliers.json using multiline option.
6. Display schema of all datasets.
7. Count records in all datasets.
8. Save raw datasets as Bronze Parquet.

Part 2: Data Cleaning

9. Find products with missing supplier_id.
10. Find inventory rows with missing stock_quantity.
11. Find sales rows with missing sale_amount.
12. Find sales rows with missing payment_mode.
13. Fill missing stock_quantity with 0.
14. Fill missing sale_amount with 0.
15. Fill missing payment_mode with Not Provided.
16. Fill missing supplier_id with UNKNOWN.
17. Create data_quality_status for products, inventory, and sales.
18. Save cleaned datasets as Silver Parquet.

Part 3: JSON Flattening

19. Flatten suppliers.json.
20. Extract contact.phone.
21. Extract contact.email.
22. Fill missing phone with Not Provided.
23. Fill missing email with Not Provided.
24. Save flattened suppliers as Parquet.

Part 4: Joins

25. Join products with suppliers.
26. Join inventory with products.
27. Join sales with stores.
28. Join sales with products.
29. Create complete retail sales DataFrame.

30. Find products with invalid supplier IDs.
31. Find inventory rows with invalid product IDs.
32. Find sales rows with invalid product IDs.
33. Find sales rows with invalid store IDs.

Part 5: Transformations

```
34. Create stock_status:
    stock_quantity <= reorder_level -> Reorder Required
    Else -> Sufficient Stock

35. Create price_category:
    unit_price >= 50000 -> Premium
    unit_price >= 10000 -> Standard
    Else -> Budget

36. Create revenue_category:
    sale_amount >= 50000 -> High Revenue
    sale_amount >= 15000 -> Medium Revenue
    Else -> Low Revenue

37. Create month column from sale_date.

38. Create year column from sale_date.

39. Create inventory_value:
    stock_quantity * unit_price

40. Create supplier_quality:
    rating >= 4.5 -> Excellent
    rating >= 4.0 -> Good
    Else -> Average
```

Part 6: Aggregations

41. Count stores by state.
42. Count products by category.
43. Count products by brand.
44. Calculate total inventory value by store.

45. Calculate total inventory value by category.
46. Count products needing reorder.
47. Calculate total revenue.
48. Calculate revenue by store.
49. Calculate revenue by city.
50. Calculate revenue by category.
51. Calculate revenue by product.
52. Calculate revenue by payment_mode.
53. Find highest revenue product.
54. Find highest revenue store.
55. Find category with highest revenue.

Part 7: Window Functions

56. Rank products by revenue.
57. Rank stores by revenue.
58. Rank products by revenue within category.
59. Find top product in each category.
60. Find top 3 products in each category.
61. Find top store by state.
62. Calculate running total revenue by sale_date.
63. Use lag to compare current day and previous day sales.
64. Use lead to show next sale amount.
65. Find products whose revenue increased compared to previous sale.

Part 8: Spark SQL

66. Create temp views for stores, products, inventory, sales, suppliers.
67. Display all sales using SQL.
68. Count products by category using SQL.
69. Calculate revenue by store using SQL.
70. Calculate revenue by city using SQL.
71. Find products needing reorder using SQL.
72. Find invalid product sales using SQL.
73. Find invalid supplier products using SQL.
74. Find top 5 products by revenue using SQL.
75. Find payment mode revenue using SQL.

Part 9: Full Refresh and Incremental Load

```
76. Write final Gold sales output using overwrite mode.
77. Partition Gold sales output by year and month.
78. Create incremental sales file for March 2026.
79. Read incremental sales file.
80. Append incremental sales to Silver sales Parquet.
81. Recalculate product revenue.
82. Recalculate store revenue.
83. Recreate Gold output partitioned by year and month.
84. Compare counts before and after incremental load.
```

Part 10: Final Gold Reports

```
85. Store Performance Report:
    store_id
    store_name
    city
    state
    total_sales
    total_revenue

86. Product Performance Report:
    product_id
    product_name
    category
    brand
    total_quantity_sold
    total_revenue

87. Inventory Reorder Report:
    store_id
    product_id
    product_name
    stock_quantity
    reorder_level
    stock_status

88. Supplier Quality Report:
    supplier_id
```

supplier_name
city
rating
supplier_quality
phone
email

89. Category Revenue Report:
    category
    total_products
    total_quantity_sold
    total_revenue

90. Payment Mode Report:
    payment_mode
    total_transactions
    total_revenue